

**Investigating the effects of spacing on working memory training outcome – a randomized controlled
multi-site trial in older adults**

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Abstract

Objective

The majority of the population will experience some cognitive decline with age. Therefore, the development of effective interventions to mitigate age-related decline is critical for older adults' cognitive functioning and their quality of life.

Methods

In our randomized controlled multi-site trial, we target participants' working memory (WM) skills, and in addition, we focus on the intervention's optimal scheduling in order to test whether and how the distribution of training sessions might impact task learning, and ultimately, transfer. Healthy older adults completed an intervention targeting either WM or general knowledge either twice per day, once per day, or once every-other-day. Before and after the intervention and 3 months after training completion, participants were tested in a variety of cognitive domains, including those representing functioning in everyday life.

Results

In contrast to our hypotheses, spacing seems to affect learning only minimally. We did observe some transfer effects, especially within the targeted cognitive domain (WM and inhibition/interference), which remained stable at the 3-month follow-up.

Discussion

Our findings have practical implications by showing that the variation in training schedule, at least within the range used here, does not seem to be a crucial element for training benefits.

Keywords: cognitive training, distributed learning, transfer

Introduction

Although only a minority of adults over age 65 will develop Alzheimer's Disease, the vast majority will experience some decline in cognitive function with age. While people vary greatly in the extent of cognitive decline, even subtle losses can substantially impact everyday life, adversely affecting critical decisions about health care, retirement, and other issues faced daily by millions of older adults (Alzheimer's Association, 2015; Tomaszewski Farias et al., 2009). Consequently, it is critically important to develop effective interventions that can contribute to the prevention of cognitive decline or improve older adults' cognitive functions, and ultimately, contribute to their well-being and quality of life.

There is growing evidence that some cognitive interventions may be effective in facilitating a range of cognitive skills with improvements evident even in healthy older adults and those with mild cognitive impairment (cf. Hill et al., 2016; Karbach & Verhaeghen, 2014; Weicker, Villringer, & Thöne-Otto, 2016 for recent meta-analyses). Some of these effects have been shown to be very long-lasting (e.g. Rebok et al., 2014). Although some controversy surrounds their real-world benefits (e.g., Simons et al., 2016), the feature shared by many promising interventions is the training focus on basic cognitive skills, often related to working memory (WM) (Karbach & Verhaeghen, 2014; Lustig, Shah, Seidler, & Reuter-Lorenz, 2009). WM is the cognitive system supporting the active maintenance of task-relevant information during the performance of a cognitive task; it is one of the main mechanisms facilitating purposeful behavior (Shah & Miyake, 1999). WM underlies performance in virtually all complex cognitive tasks and critically, it is highly susceptible to age-related decline (Bugg, Zook, DeLosh, Davalos, & Davis, 2006; Park et al., 2002) and predictive for everyday functioning in old age (Cahn-Weiner et al., 2007; Tomaszewski Farias et al., 2009). Furthermore, experimental and neuroimaging studies have demonstrated a close relationship between WM and episodic memory (e.g., Buckner, 2004; Flegal & Reuter-Lorenz, 2014). Corroborating those findings, several intervention studies observed improvements in various measures of episodic memory after WM training (e.g., Buschkuehl et al., 2008; Flegal, Ragland, & Ranganath, 2019; Richmond, Morrison, Chein, & Olson, 2011), which might be driven by common neural networks between the two cognitive domains (Dahlin, Neely, Larsson, Backman, & Nyberg, 2008).

Despite promising demonstrations that WM training is effective in older adults (e.g., Borella, Carretti, Zanoni, Zavagnin, & De Beni, 2013; Stepankova et al., 2014), several studies report only minimal transfer (e.g., Brehmer et al., 2011; Richmond et al., 2011), whereas others report effects in young adults with more limited or no effects in older adults using the same intervention (e.g., Brehmer, Westerberg, & Bäckman, 2012; Dahlin et al., 2008). Such findings suggest that some current interventions may not meet the needs of older adults, that age-related limitations in plasticity are difficult to overcome, or both. As such, more research is needed to make WM interventions more suitable for older adults, and also, to make them more robust by determining training features that might moderate transfer in an aging population. Specifically, systematic research is needed to determine the optimal scheduling of the training sessions, i.e., the effect of spacing to maximize learning outcomes. Despite the extensive literature on spaced vs massed training in the domains of skill and verbal learning and their enormous practical significance (e.g., Cepeda et al., 2009; Krug, Davis, & Glover, 1990), spacing effects have been rarely tested in the context of WM training. To our knowledge, there is one study to date demonstrating that spaced and massed practice produced minimal differences in specific training effects, yet distributed practice had a significant impact on immediate transfer (post-test performance). Specifically, the group showing the largest transfer effects completed one training session per day (as opposed to the groups that trained multiple times per day) (Wang, Zhou, & Shah, 2014), which is consistent with a host of prior results demonstrating that distributed practice leads to better learning and retention than massed practice (Cepeda et al., 2009). Because this study was conducted in children, it is unclear whether and how the results might translate to an older population.

To address these outstanding questions in the literature, the present study investigates the spacing of training sessions and how different spacing schedules might affect training performance and transfer in an older adult population. We focus on long-term outcomes given that the effects of spacing have often been observed after a longer delay (Cepeda, Pashler, Vul, Wixted, & Rohrer, 2006). The primary goal of the current study was to test whether or not the training regimen impacted WM and related processes such as inhibitory control. Given previous reports, we also tested whether training WM might transfer to measures of episodic memory,

especially those that serve as proxy for every-day memory functions and problem solving (Cantarella, Borella, Carretti, Kliegel, & Beni, 2017; Weicker et al., 2016). Specifically, we conducted a randomized controlled multi-site trial, in which a relatively large sample of older adult participants was assigned to either an intervention targeting WM, or an alternative intervention that focused on general knowledge and vocabulary learning. All participants were instructed to complete 20 sessions of training at home using a tablet device. Within both interventions, participants were further assigned to complete their sessions either twice per day, once per day, or once every other day. Before and after the intervention, as well as three months after training completion, participants' cognitive functions in various domains were tested in order to assess potential transfer effects.

Based on the previous literature, we predicted that the WM-training group would outperform the knowledge-training group in measures closely related to the trained task, specifically, in WM and interference/inhibitory control. We also explored potential improvements in long-term memory measures, especially those that are proxies for real-world performance. We did not expect improvements in basic processing speed and vocabulary, which served as control measures; if anything, the knowledge-training group should show more pronounced improvement in the vocabulary measure as compared to the WM-training group given that this was the focus of the alternative intervention.

Within the WM-training group, we predicted that the every-other-day group would outperform the twice-per day group (and potentially, the every-day group) during training, and especially during the long-term follow-up sessions for which we were expecting the effects of spacing to become most apparent (Cepeda et al., 2006). We also expected the effects of spacing to be most pronounced in the measures that are most closely related to the trained task.

Methods

Participants and Design: After passing an initial phone screening to establish eligibility criteria, 183 participants were randomly assigned to either the WM- or knowledge-training group. Furthermore, within each

group, participants were randomly assigned to one of three spacing conditions that required them to complete their intervention either twice per day, once per day, or once every other day (see Figures 1 and S1 in the Supplementary Online Material (SOM)). After intervention completion, participants returned for the post-test, as well as for a follow-up assessment 3 months later. Fifty-seven percent of the participants were recruited in Southern California, and 43% were recruited in Southeast Michigan. All research procedures were approved by an Institutional Review Board and participants signed an informed consent.

Figure 1 about here.

Demographic variables for the intervention groups, and for those who did not complete the study or were excluded, as well as further participant details are provided in the SOM (p. 2 and Table S1). The two intervention groups did not differ significantly in any of the demographic variables. Importantly, the individuals who were excluded or dropped out of the interventions were not different from the analyzed sample, except for self-reported socioeconomic status (SES; the analyzed sample reported lower SES), and the number of completed training sessions (the analyzed sample completed more sessions; see Table S1).

Tasks and Materials:

Training Tasks:

Working Memory Training. Participants trained on a tablet-based version of the n-back task that used pictures as stimuli, and required indicating whether a presented picture was the same as the one presented n trials previously (similar to the task used as outcome measure in Katz, Jaeggi, Buschkuhl, Stegman, & Shah, 2014). The stimuli were presented in a moving window that lasted for 1,000ms with an inter-stimulus interval of 2,500ms. Each stimulus could be a target trial, a non-target trial, or a lure trial (i.e., stimuli that were the same as the target stimuli, except that they were presented in the wrong position, such as $n \pm 1$ back). The task was adaptive in that the level of difficulty (number of $n-1/n+1$ lures and n-back level) changed with respect to participants' performance after each round (Jaeggi, Buschkuhl, Shah, & Jonides, 2014). Participants worked on the same n-back level with no lures, few lures (i.e., 2), and many lures (i.e., 6), and moved up an n-back level after clearing the many-lures round or moved down an n-back level after making too many errors in the no-lure round. Participants completed 10 rounds per training session, and each round consisted of 5 target trials, 10+n

non-targets trials, a variable number of lures (0, 2, or 6), and either 0, 2, or 6 filler trials at the start of the round, depending on what lure condition was presented. The dependent variable was the average level of *n* reached per training session.

Knowledge Skills Training. Participants used a tablet-based general knowledge task as described previously (Jaeggi et al., 2014). Participants were presented with GRE-type general knowledge, and vocabulary questions, along with four answer alternatives. After participants selected their response, they received feedback, and questions answered incorrectly were presented again in the beginning of the next session in order to promote learning. The task was adaptive in that the difficulty of the questions was reflected in levels as determined by pilot testing. Although the emphasis was on accuracy, participants were given a time limit of 45 seconds to respond before a trial was automatically marked as incorrect. The dependent variable was the average level of difficulty achieved in each training session.

Baseline / Control Assessments:

We administered a detailed *demographic and health questionnaire* to assess participants' general physical and psychological health and well-being, including the *WHOQOL-OLD* (Fang et al., 2012). We also screened for general cognitive status using the *Mini-Mental State Examination* (MMSE; Tombaugh & McIntyre, 1992), *Geriatric Depression Scale* (GDS; Yesavage, 1988), and *Generalized Anxiety Depression Questionnaire* (GAD-7; Spitzer, Kroenke, Williams, & Löwe, 2006) (cf. Table S1; SOM).

The following measures were administered at baseline, as well as during post- and follow-up sessions, although they were considered control measures, for which we did not expect any changes:

Vocabulary

To assess general intellectual ability, we used the *Mill Hill Vocabulary test* (Raven, Raven, & Court, 1998) in which 33 target words were presented with 6 potential synonyms, and participants had to select the appropriate word that matches the target word. The dependent variable was the number of correctly identified synonyms. We used parallel-test versions counterbalanced across participants and sessions.

Processing Speed

Pattern and Letter Comparison (Ribaupierre & Lecerf, 2006). We administered two variants of paper-pencil perceptual and motor speed tasks in which participants had to indicate as quickly as possible whether two patterns or letter strings presented next to each other are the same or not (e.g., QLXVST __ QLNSVT). There were 60 items for the pattern comparison, and 42 items for the letter comparison, and we used the total time to complete each of the tasks (in seconds) as the dependent variable.

Transfer Measures

In order to test for *training-related changes* (i.e., *transfer*), we administered the following set of tasks at baseline, as well as during the post- and follow-up test.

Working Memory

Spatial N-back. As a measure of near transfer, we administered a spatial version of the trained N-back task, also via tablet devices. The task parameters were the same as for the training task except that the task was not adaptive; that is, after one round of 1-back, the task difficulty was set to a 2-back level. Participants completed 3 rounds of 2-back without lures and 3 rounds of 2-back with many (6) lures. The dependent variable was *pr* (proportion hits minus false alarms) as well as reaction time (median) for correct trials, averaged across all 2-back trials. In addition, we used the false alarm rate separately to specifically capture aspects of interference resolution and memory precision.

Sternberg Task. We used a similar version of a computerized item-recognition task as used in Iordan et al. (2018). Participants were required to encode and retain a set of consonant letters in uppercase (set size 4-8) for several seconds, and after a brief retention interval, they were given a probe letter in lowercase and were asked to indicate whether or not this probe letter was part of the initial memory set. Participants completed 3 blocks of 20 trials each (4 trials per set size), and the dependent variables were accuracy and reaction time (median) for correct responses across all trials.

Symmetry Span. This computerized task was adapted from Redick et al. (2012). Participants had to indicate whether or not a pattern was symmetrical, after which a square was presented in one out of 16 locations on a grid. After 2-6 trials (symmetry decision + location), participants were asked to recall the locations in order

using the computer mouse. We used parallel-test versions counterbalanced across participants and sessions, and the number of correctly recalled sets served as the dependent variable.

Inhibitory Control / Interference

D2 (Brickenkamp, 2002). We administered a paper-pencil version of the D2 which consists of 14 lines of letters (either p or d) with 1-4 dashes below and/or above each letter. Participants are provided 20 seconds per line and asked to cross out any ds with two dashes as quickly as possible while ignoring all other items. The total number of items completed minus any type of error (commission or omission; TN-E) was our index of inhibitory control.

Episodic Memory

Visual LTM. This paper-pencil task was adapted from Perrig et al. (2006). Participants were shown two arrays of Snodgrass & Vanderwart-like line-drawings of objects, patterns and words on a single page, and they had to mark as many differences as possible between the two arrays within 3 minutes. About 20 minutes later, participants recalled all they could remember from the picture, as well as the differences they found. The dependent variables were the number of correctly recalled items. In addition, in order to capture another aspect of interference and memory precision, we used intrusions (incorrectly recalled items). We used parallel-test versions counterbalanced across participants and sessions.

Meta Memory (MM). This task was adapted from McGillivray and Castel (2011), cf. Parlett-Pelleriti et al. (2019). Participants were presented with five 12-word lists, and after each word, they were asked to place a bet between 0 and 10 points representing the likelihood of remembering that word later on. At the end of each list, participants were asked to recall as many words as possible. For each correctly remembered word, the bet for that word was added to their score, and for failure to recall a word, it was subtracted from their score. After each list, participants were presented with their score before proceeding to the next list. Parallel-test versions were administered, counterbalanced across participants and sessions. The number of correctly recalled words across all lists served as the dependent variable.

Characterization of the Elderly on Daily Activities in the Real-world (CEDAR; Everyday Memory Test).

This verbal prompting task was adapted from Thomas (2015) to assess the performance of everyday cognition

akin to the Observed Tasks of Daily Living (Diehl et al., 2005). Participants were assigned the role of a neighbor for a fictitious character and asked by a fictitious relative to complete a series of errands that involved managing medications, meal planning, finances, and making long-term decisions (e.g., bank/doctor selection). A parallel version was used, counterbalanced across participants and sessions. We used accuracy (standardized across subtasks and averaged into one measure) as the dependent variable.

Analytical Approach

Using SPSS 24, JASP 0.9.1.0, and R, we conducted analyses to investigate the effects of spacing on training performance, and potentially, generalizing effects to non-trained outcome measures (cf. SOM for further details). Our hypotheses and analytical approach have been pre-registered (cf. AsPredicted #7897; <https://aspredicted.org/mp2jv.pdf>).

Results

Specific training effects

The performance of the WM training group improved over the course of training, from an average n-back level of 2.36 (SD = .41) obtained across the first two sessions to an average of 2.81 (SD = .74) in the last two sessions ($t(77) = 6.29, p < .001; d = .72; BF_{10} = 1.342e+6$).

However, the spacing condition had no effect on training, as indicated by a repeated-measures ANOVA with spacing condition as the between-subject factor and session number (1-20) as the within-subject factor (main effect of spacing condition: $F(2,58) = .52; p = .60; \eta^2_p = .02; BF_{10} = .29$; spacing x session interaction: $F(38,1102) = .90; p = .64; \eta^2_p = .03; BF_{10} = .001$), or with a univariate ANOVA using gain (calculated as the difference between the performance during the two last sessions and the first two sessions) as the dependent variable ($F(2,75) = .24; p = .79; \eta^2_p = .006; BF_{10} = .13$), cf. Figure 2a and Table 1.

The performance of the knowledge-training group also improved, from an average difficulty level of 5.33 (SD = 1.27) across the first two sessions to an average of 7.01 (SD = 2.08) across the last two sessions ($t(76) = 12.32, p < .001; d = 1.41; BF_{10} = 1.198e+17$). Again, spacing condition had no effects on training

(repeated-measures ANOVA: main effect of spacing: $F(2,59)=2.10$; $p = .13$; $\eta^2_p = .07$; $BF_{10} = .87$; spacing x session interaction: $F(38,1121)=1.10$; $p = .32$; $\eta^2_p = .04$; $BF_{10} = .006$; univariate ANOVA (gain): $F(2,74) = .13$; $p = .88$; $\eta^2_p = .003$; $BF_{10} = .12$), cf. Figure 2b and Table 1.

Figure 2 about here

However, across both groups, there was a difference in the number of completed training sessions in that the twice-per-day group completed significantly more training sessions on average than the every-other-day group ($M=21.45$ ($SD=3.63$) vs. $M=19.90$ ($SD=2.03$); $t(104)=2.56$; $p = .007$; $d = .56$; $BF_{10} = 4.76$).

To investigate differential training effects as a function of spacing conditions, we calculated univariate ANOVAs separately for each intervention group using participants' final performance (average last two sessions) as well as the gain as the dependent variables (cf. Table 1). In addition, we conducted individual curve-fitting analyses using linear as well as spline regression approaches to analyze early vs late learning (Jordan et al., 2018; Lövdén, Bäckman, Lindenberger, Schaefer, & Schmiedek, 2010) (cf. SOM). These analyses also failed to reveal any group differences due to spacing condition. The only hint of a spacing effect was a main effect of spacing condition in the change from early to late learning for the WM-training group: for the every-day group the improvement was more pronounced compared to the every-other-day group ($p = .03$; $d = .67$; $BF_{10} = 2.42$). However, no group differences were evident in any of the other outcome measures, indicating that overall, spacing effects on training were negligible for both the WM- and the knowledge-training group.

Table 1 about here

Note that there was some variability in adherence to the prescribed training schedule for various reasons (e.g., forgetting to train, travel, illness, catching up with missed training sessions, etc.). Thus, we conducted additional analyses on only those participants who strictly followed the assigned schedule across all 20 training sessions. Once again, no effects of spacing emerged for any of the training measures (see Figures 3a and 3b, and Table S2; SOM).

Transfer effects

Descriptive information for each of the measures as a function of group and testing session is provided in Tables S3 and S4 (SOM). There were no group differences in performance at pre-test for any of the variables (all p s > .11; all BFs < .18).

A comprehensive multivariate analysis of variance (MANOVA) using gain scores (pre vs. post, and pre vs. follow-up) that included all outcome measures (accuracy) for which we expected transfer resulted in overall significant effects of intervention in favor of the WM-training group at both, post-test and follow-up (*pre vs. post*: $F(9,125)=3.30$, $p=.001$, $\eta^2_p=.19$; *pre vs. follow-up*: $F(9,81)=2.48$, $p=.015$, $\eta^2_p=.22$). Adding spacing group as additional variable did not change the results, and neither the main effects of spacing group, nor the intervention x spacing condition interactions were significant (all p > .06).

Furthermore, we calculated analyses of covariance (ANCOVA) using post-test/follow-up performance as the dependent variable, group (WM vs knowledge training) as a between-subjects factor, and pre-test performance as a covariate. Spacing condition was included as an additional between-subjects factor. The results for the various cognitive domains are reported in the main text below, and in addition, the results for each individual measure are provided in Tables 2 and 3¹.

A confirmatory factor analysis revealed that a 3-factor model represented our data adequately (see SOM; Figure S4a), and thus, we tested for intervention effects with ANCOVAs using the composites (z-scored) of the post-tests as dependent variables and pre-tests (z-scored composite) as covariates for each of the three constructs (WM, inhibition/interference, episodic memory)². In addition, we conducted separate analyses for the

¹ In the SOM, we also report the results of MANOVAs that include the gain scores for each of the measures within a construct.

² Note that a 2-factor model where the WM and inhibition/interference measures were combined into one construct represented our data equally well (cf. SOM, Figure S4b for details).

WM reaction time (RT) measures (n-back and Sternberg), and control measures (processing speed, vocabulary). See SOM for more details.

For the WM accuracy composite, the WM-training group outperformed the knowledge-training group at post-test ($F(1, 141) = 3.98, p = .02; \eta^2_p = .03; BF_{10} = 1.05$), a small effect that was reduced at follow-up ($F(1, 95) = 2.29, p = .07; \eta^2_p = .02; BF_{10} = .56$). Although there was an overall effect on the composite at post-test, this was driven mostly by the n-back task, which showed strong effects in both post and follow-up sessions (cf. Tables 2 and 3). In the WM RT composite, the WM-training group similarly outperformed the knowledge-training group (post: $F(1, 146) = 7.33, p = .004; \eta^2_p = .05; BF_{10} = 5.06$), and this effect, though small, was sustained at follow-up ($F(1, 99) = 3.07, p = .04; \eta^2_p = .03; BF_{10} = .22$). The improvement here was mostly driven by the Sternberg task, which - by itself - showed particularly strong effects at the follow-up test (cf. Tables 2 and 3). The addition of spacing condition as a between-subject factor in the ANCOVA did not change any of the results, and moreover, there were no main effects of spacing or intervention x spacing interactions in either of the composite measures (all $ps > .26$; all $BF_{10}s < .09$).

For the inhibition/interference composite, the WM-training group outperformed the knowledge-training group at post-test, ($F(1, 145) = 4.12, p = .02; \eta^2_p = .03; BF_{10} = 1.18$), controlling for pre-test performance, and the effect was similar at follow-up ($F(1, 98) = 4.96, p = .01; \eta^2_p = .05; BF_{10} = 1.08$), although it was small in both cases. The improvements at post-test were mainly driven by the n-back false alarm rate, but at follow-up, participants in the WM-training group also outperformed the knowledge-training group in the D2 measure to some extent (cf. Tables 2 and 3). Again, adding spacing condition as a factor in the analyses did not change the results, and there were no effects of spacing (all $ps > .17$; all $BF_{10}s < .13$).³

³ When using the combined WM/interference composite (accuracy; Figure S4b), the WM-training group also outperformed the knowledge-training group at post-test controlling for pre-test performance ($F(1, 137) = 6.81, p = .005; \eta^2_p = .05; BF_{10} = 3.73$), and the effect was similar at follow-up ($F(1, 93) = 4.49, p = .02; \eta^2_p =$

In the episodic memory composite, there were no group differences at post-test, controlling for pre-test performance ($F(1, 142) = .04, p = .42; \eta^2_p < .001; BF_{10} = .18$), and neither was there an effect at follow-up ($F(1, 95) = .63, p = .21; \eta^2_p = .007; BF_{10} = .28$). Again, adding spacing condition to the analyses did not change the results, and there were no effects of spacing either (all $ps > .23$; all $BF_{10}s < .04$).

As predicted, there were no effects for the control measures favoring the WM- or knowledge-training group (*vocabulary*: post: $F(1, 150) = 2.98, p = .09; \eta^2_p = .02; BF_{10} = 1.09$; follow-up: $F(1, 99) = 1.64, p = .20; \eta^2_p = .02; BF_{10} = .57$; *processing speed*: $F(1, 149) = 1.41, p = .24; \eta^2_p = .009; BF_{10} = .45$; follow-up: $F(1, 104) = 1.62, p = .21; \eta^2_p = .02; BF_{10} = .40$). Adding spacing condition to the analyses did not change the results, and there were no effects of spacing either (all $ps > .26$; all $BF_{10}s < .15$).

Tables 2 and 3 about here

Discussion

In this study, we tested whether the distribution of training sessions, i.e., the spacing of training, plays a role in cognitive training performance and/or transfer in a healthy older adult population. Using three different training schedules (training twice per day, once every day, or once every other day), participants completed either an intervention targeting WM, or an alternative intervention that required fact and vocabulary learning over the course of 20 sessions. Our data provide little evidence that spacing plays any role in learning, at least with the fairly restricted spacing schedule, or the training tasks and outcome measures used here, neither when using the full sample, nor with a sample limited only to the participants who adhered strictly to the assigned spacing protocol.

Although descriptively, the participants in the every-other-day condition seem to underperform during training compared to the other two spacing conditions within the WM-training group (Figure 2a), potentially

.05; $BF_{10} = 1.43$). The addition of spacing condition as a factor in the analyses did not change the results, and there were no effects of spacing (all $ps > .27$; all $BF_{10}s < .14$).

reflecting age-related deficits in consolidation (Spencer, Gouw, & Ivry, 2007), we did not detect any reliable differences, at least with the measures and statistical approaches used here. Furthermore, those differences disappear in the reduced sample that strictly adhered to the spacing protocol (Figure 2c), suggesting that factors other than spacing might be at play there.

In terms of transfer, overall, we did observe effects in favor of the WM-training group immediately after training completion, as well as 3 months after training completion as evidenced by a comprehensive MANOVA that included all accuracy measures. When using theoretically derived composites, we observed that the improvements were driven by non-trained measures of WM and inhibition/interference, and thus, our data are consistent with previous work in older adults (Borella et al., 2013; Stepankova et al., 2014). We observed transfer to WM using both a composite measure representing accuracy and another that represented reaction times. While accuracy effects were exclusively driven by the WM-training group's improvements in a non-trained spatial variant of the n-back task, the improvements in reaction time were driven by the WM-training group's performance in a Sternberg item recognition task. Importantly, the improvements in those individual measures remained present at the 3-month follow-up, indicating that the effects are not simply short-lived. In addition to the observed transfer to WM functions, the WM-training group demonstrated improvements in a composite representing inhibition/interference, which were mainly driven by reduced false alarm rates in the non-trained variant of n-back, but also by a more pronounced improvement in the D2 measure, especially at follow-up. Given that our intervention deliberately incorporated the requirement to resolve interference through lure trials, the transfer effects observed in the n-back task might indicate improvements in memory precision, whereas the improvements in D2 are more indicative of changes in inhibitory control.

The transfer effects we observed in WM and inhibition/interference are consistent with our previous neuroimaging work in young adults demonstrating that WM measures that also include an interference component share overlapping neural networks (Hsu, Jaeggi, & Novick, 2017), and as such, our observed transfer effects along with the finding that the WM and inhibition/interference measures can be represented by the same latent construct provide behavioral evidence for that notion.

In contrast to previous work, however, there was no evidence of transfer to measures of episodic memory. Although we did observe improvements in a visual long-term memory measure successfully used in previous intervention work (Buschkuhl et al., 2008), in the present study, both the WM- and knowledge-training groups improved to a similar extent. Thus, despite the relatively large effect sizes in both groups, it is not clear whether those improvements go beyond test-retest effects due to the fact that we do not have a no-contact control group. No improvements were evident in either of the other episodic memory measures, including the one that represented everyday memory functioning (CEDAR). Thus, we conclude that the n-back intervention used here might not be an ideal vehicle for improving such skills and that other approaches, such as those that include a focus on meta-cognitive strategies, might be more promising (e.g. Strickland-Hughes, 2017; Vranic, Spanic, Carretti, & Borella, 2013).

Several limitations of this study should be noted. Specifically, due to the design of the study, the participants in the three spacing conditions completed their pre- and post-test assessments over different intervals. That is, while it took the twice-per-day group only two weeks to complete the intervention, the every-other-day group took three times as long. This difference may contribute to the lack of spacing effects in the transfer measures because memory effects due to re-testing might be stronger in groups with shorter intervals between assessments, and as such, any benefits of spacing may have been overridden by the time between pre-test and post-test. However, the pre-post time interval should be less relevant at the 3-month follow-up, where spacing effects were expected to be larger, yet none were observed for any measures. Nonetheless, a design that controls for the timing between training and assessment sessions could shed more light on this issue.

Another limitation is the restricted range of spacing conditions used here. More extreme spacing conditions (e.g., conditions that require completing all 20 sessions within two days, vs completing only one session per week) might have led to performance differences that better reflect massed vs distributed learning. Such a design has its own logistical challenges though, including the issue of time between assessments mentioned above. Alternatively, combining massed and spaced approaches within one group might be beneficial, especially in an older population. Specifically, the first few sessions could be held in a massed protocol in order to maximize task familiarization and early learning (Lövdén et al., 2010), and over time, the

sessions could be increasingly spaced to gauge the actual effects of distributed learning. Such an approach has intuitive appeal; however, the relevant literature is limited, conflicting (Goedert & Miller, 2008; Hauptmann, Reinhart, Brandt, & Karni, 2005), and has not examined the types of tasks trained here.

Nonetheless, it is important to recognize that our results might only apply to the specific form of cognitive training used here (WM and knowledge training), and that the lack of measurable spacing effects is restricted to a relatively narrow range of spacing conditions. It is critical for future research to test different variations of training schedules using other interventions (e.g., speed of processing training) over longer periods of time with appropriate outcome measures, and to replicate the present results with populations that differ in demographic and health characteristics (e.g., young adults).

Nonetheless, our results have practical implications. Even though our older participants were asked to perform a demanding cognitive intervention unsupervised and independently at home, they were not only able to complete their intervention and showed considerable task-specific improvements, but also, they were reasonably compliant. Almost everyone who started the intervention completed the requested number of training sessions. In fact, many participants (especially those in the twice-per-day condition) trained more than the required amount, some training more than three times the required sessions, resulting in their exclusion from the analyses (cf. Figure 1). Our data reflect what participants might be doing in the real-world, e.g., when they choose to complete cognitive training on their own. It seems that even complex interventions do not necessarily have to be conducted in the lab following a precise training schedule in order to be effective.

Conclusions and Implications

Our study demonstrated that healthy older adults are able to complete a fairly complex intervention on tablet devices independently at home with minimal researcher interaction (see also Stepankova et al., 2014). Furthermore, training WM seems to have some benefits in terms of improving performance in non-trained measures, in particular, WM and inhibition/interference, which are sustained up to three months after training completion. However, in contrast to other studies, our intervention did not result in improvements in episodic memory including a measure that reflects real-world requirements.

Most critically, our results indicate that the spacing of training sessions seems to have negligible effects on training outcome, at least within the restricted range of spacing conditions used here, and that other factors may be more important for producing more robust training and transfer outcomes.

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Table 1. Training data as a function of intervention group and spacing condition

	Every Other		Every Day		Twice per		Main Effect		
	Day				Day		(Spacing)		
	Mean	SD	Mean	SD	Mean	SD	<i>p</i>	<i>BF</i>	η^2p
Working Memory	N=26		N=24		N=28				
Training Group									
Number of sessions	19.46	2.14	20.46	2.13	21.61	3.78			
Range of sessions	14-26		17-26		14-29				
Average first 2 sessions	2.25	.45	2.39	.35	2.44	.42	.24	.35	.04
Average last 2 sessions	2.65	.53	2.91	1.10	2.88	.50	.39	.23	.03
Overall gain	.39	.41	.52	.97	.44	.41	.79	.13	.01
<i>Linear Regression</i>									
Overall slope	.02	.02	.03	.05	.01	.02	.37	.24	.03
Intercept	2.39	.47	2.57	.48	2.64	.39	.13	.58	.05
<i>Spline Regression</i>									
First slope	.21	.25	.40	.43	.27	.24	.09	.73	.06
Change from first slope	-.18	.29	-.41	.39	-.29	.27	.05	1.20	.08
Second slope	.03	.09	.00	.07	-.01	.13	.35	.26	.03
Knot location	6.20	3.91	4.75	3.14	5.48	3.74	.38	.24	.03
Knowledge Training									
Group	N=25		N=24		N=28				
Number of sessions	20.36	1.85	19.88	1.70	21.29	3.54			
Range of sessions	15-25		16-26		13-30				
Average first 2 sessions	5.31	1.10	5.14	1.38	5.50	1.32	.61	.17	.01
Average last 2 sessions	7.07	1.83	6.74	2.25	7.18	2.19	.73	.14	.01

Overall gain	1.77	1.27	1.59	1.10	1.68	1.25	.88	.12	.00
<i>Linear Regression</i>									
Overall slope	.08	.06	.07	.05	.06	.05	.56	.18	.02
Intercept	5.85	1.37	5.57	1.86	6.14	1.86	.49	.20	.02
<i>Spline Regression</i>									
First slope	1.54	1.39	1.45	1.87	1.88	1.54	.61	.17	.01
Change from first slope	-1.61	1.38	-1.47	1.88	-1.77	1.68	.81	.13	.01
Second slope	-.07	.45	-.02	.26	.10	.29	.18	.43	.05
Knot location	5.12	4.07	5.63	4.19	4.37	3.68	.53	.19	.02

Note. BF = Bayes Factors.

Table 2. Intervention effects at posttest (ANCOVAs; pre-test performance as covariate)

Outcome Measure	df	F	p	η^2p	BF
<i>Working Memory Measures</i>					
Spatial N-back (Acc)#	1, 150	29.97	***	.17	63000.43
Sternberg (Acc)	1, 149	.10	.76	.00	.18
Symmetry Span (Acc)	1, 145	.03	.88	.00	.18
Spatial N-back (RT)#	1, 150	.71	.40	.01	.12
Sternberg (RT)	1, 149	8.09	**	.05	6.57
<i>Inhibitory Control Measures</i>					
Spatial N-back (False Alarms)	1, 149	10.87	***	.07	21.85
D2 (TN minus E)	1, 151	1.47	.23	.01	.35
VLTM (Intrusions)	1, 149	.20	.66	.00	.18
<i>Long-Term Memory Measures</i>					
Meta-Memory (Recall)	1, 146	.01	.94	.00	.18
VLTM (Recall)	1, 149	.28	.60	.00	.20
CEDAR (Acc)	1, 151	.63	.43	.00	.23
<i>Control Measures</i>					
Mill-Hill Vocabulary	1, 150	2.98	.09	.02	.70
Letter Comparison	1, 149	.45	.50	.00	.21
Pattern Comparison	1, 151	1.35	.25	.01	.31

Note. p-values and Bayes Factors (BF) are 2-tailed (uncorrected). BF above 1 (in bold) are considered evidence in favor of the alternative hypothesis. # = assumptions for ANCOVA were not met, thus, the values for the session x group interaction (repeated measures ANOVA) are reported instead. **p<.01; ***p<.001 (2-tailed).

Table 3. Intervention effects at follow-up (ANCOVAs; pre-test performance as covariate)

Outcome Measure	df	F	p	η^2p	BF
Working Memory Measures					
Spatial N-back (Acc)	1, 101	15.51	***	.13	144.68
Sternberg (Acc)	1, 102	.00	.95	.00	.21
Symmetry Span (Acc)	1, 99	.00	.96	.00	.21
Spatial N-back (RT, correct)	1, 101	.04	.84	.00	.21
Sternberg (RT, correct)	1, 102	10.54	**	.09	19.87
Inhibitory Control Measures					
Spatial N-back (False Alarms)	1, 101	10.19	**	.09	16.01
D2 (TN minus E)	1, 101	3.49	.07	.03	.97
VLTM (Intrusions)	1, 103	.02	.88	.00	.21
Long-Term Memory Measures					
Meta-Memory (Recall)	1, 97	.00	.97	.00	.22
VLTM (Recall)	1, 103	.08	.78	.00	.22
CEDAR (Acc)	1, 103	.53	.47	.01	.26
Control Measures					
Mill-Hill Vocabulary#	1, 99	.38	.54	.00	.25
Letter Comparison	1, 104	.89	.35	.01	.30
Pattern Comparison	1, 104	2.49	.12	.02	.62

Note. p-values and Bayes Factors (BF) are 2-tailed (uncorrected). BF above 1 (in bold) are considered evidence in favor of the alternative hypothesis. # = assumptions for ANCOVA were not met, thus, the values for the session x group interaction (repeated measures ANOVA) are reported instead. **p<.01; ***p<.001 (2-tailed).

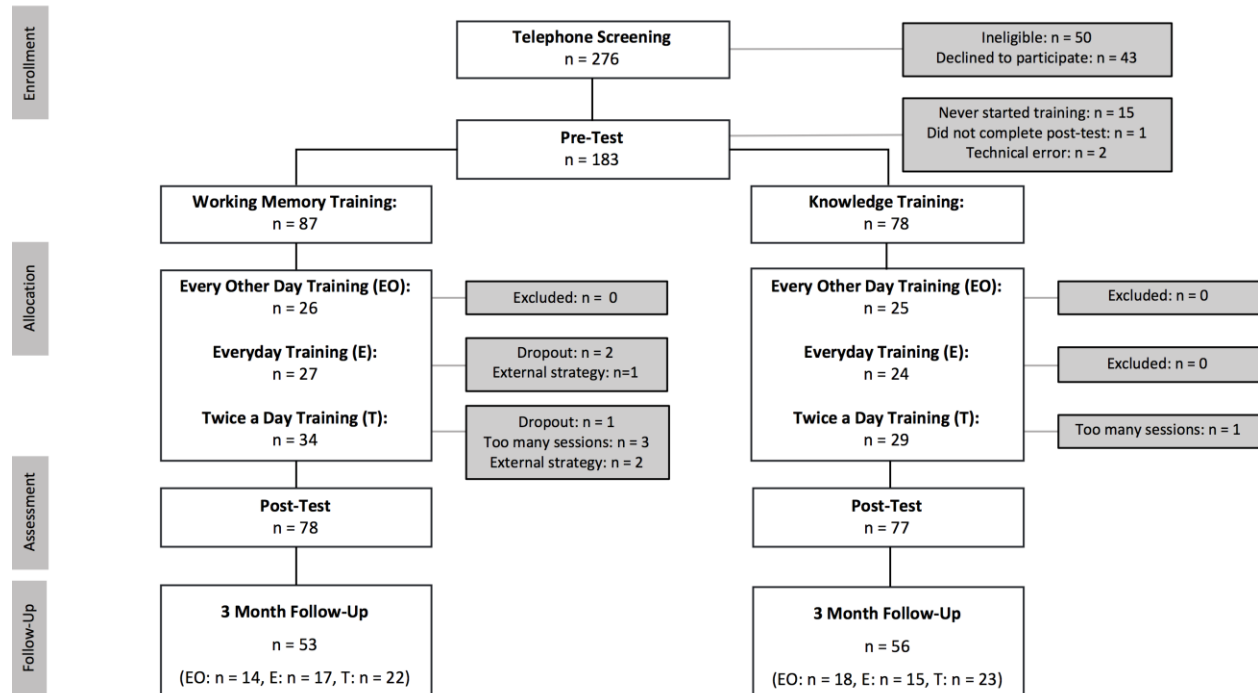
Figure Captions

Figure 1. Flowchart of participants and allocation.

Figure 2. Training performance as a function of spacing condition and intervention group. a) Working memory-training group. b) Knowledge-training group. Error bars represent standard errors of the mean.

Figure 3. Training performance as a function of spacing condition and intervention group in the restricted sample that includes only participants who followed the spacing schedule as defined by using a conservative classification approach (cf. SOM). a) Working memory-training group. b) Knowledge-training group. Error bars represent standard errors of the mean.

Figure 1



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Figure 2a

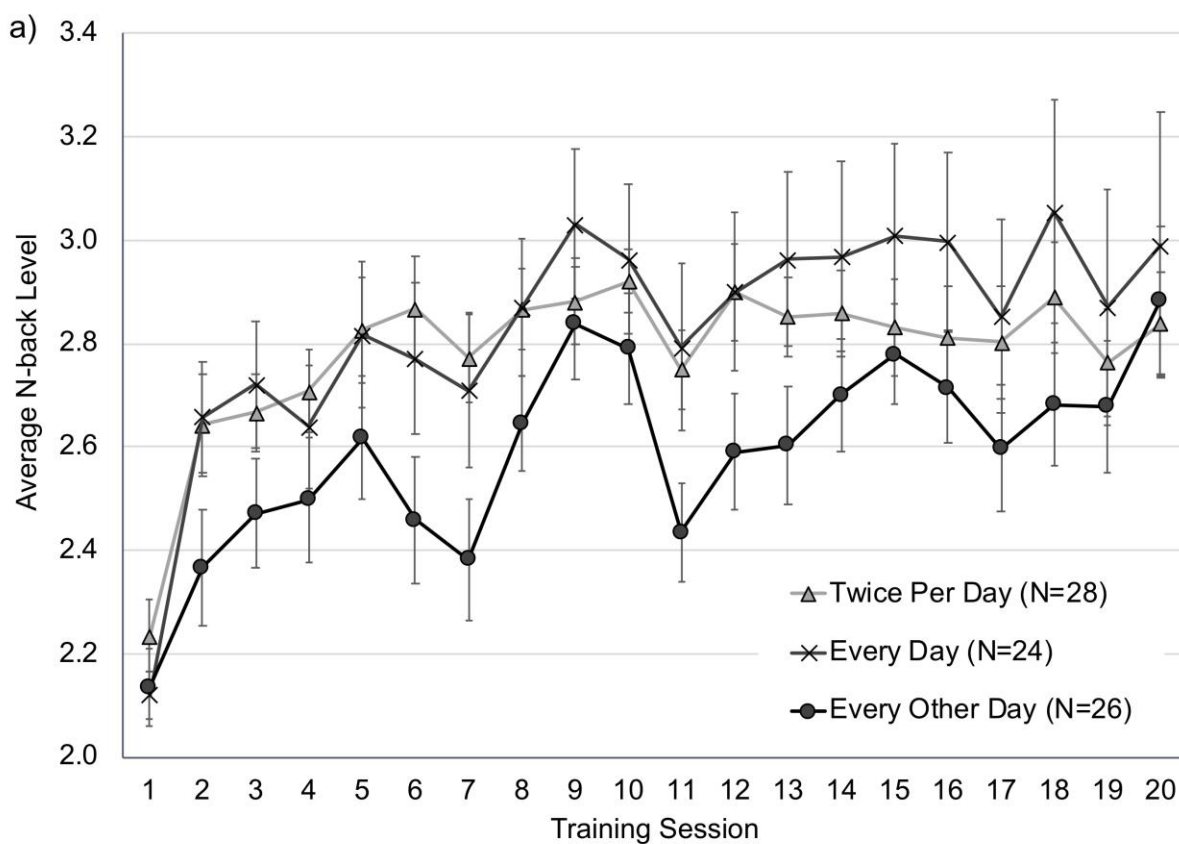


Figure 2b

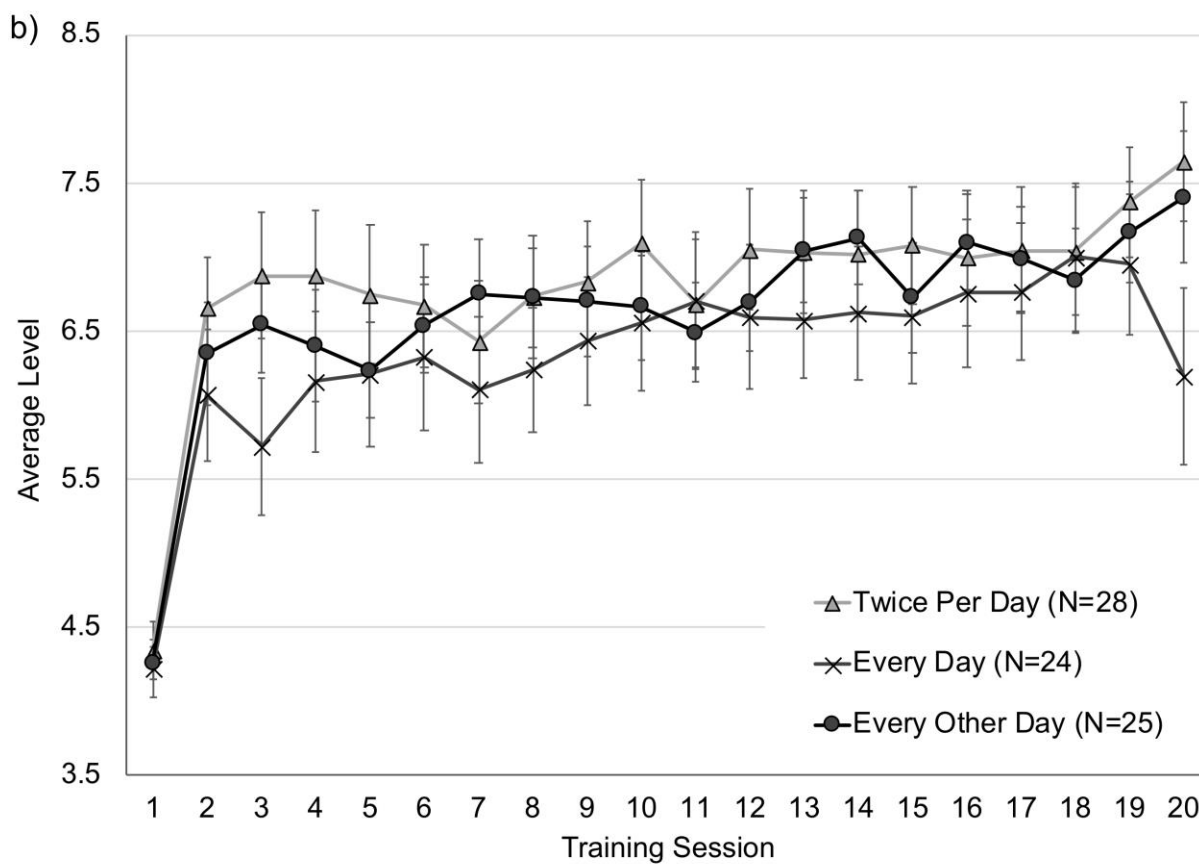


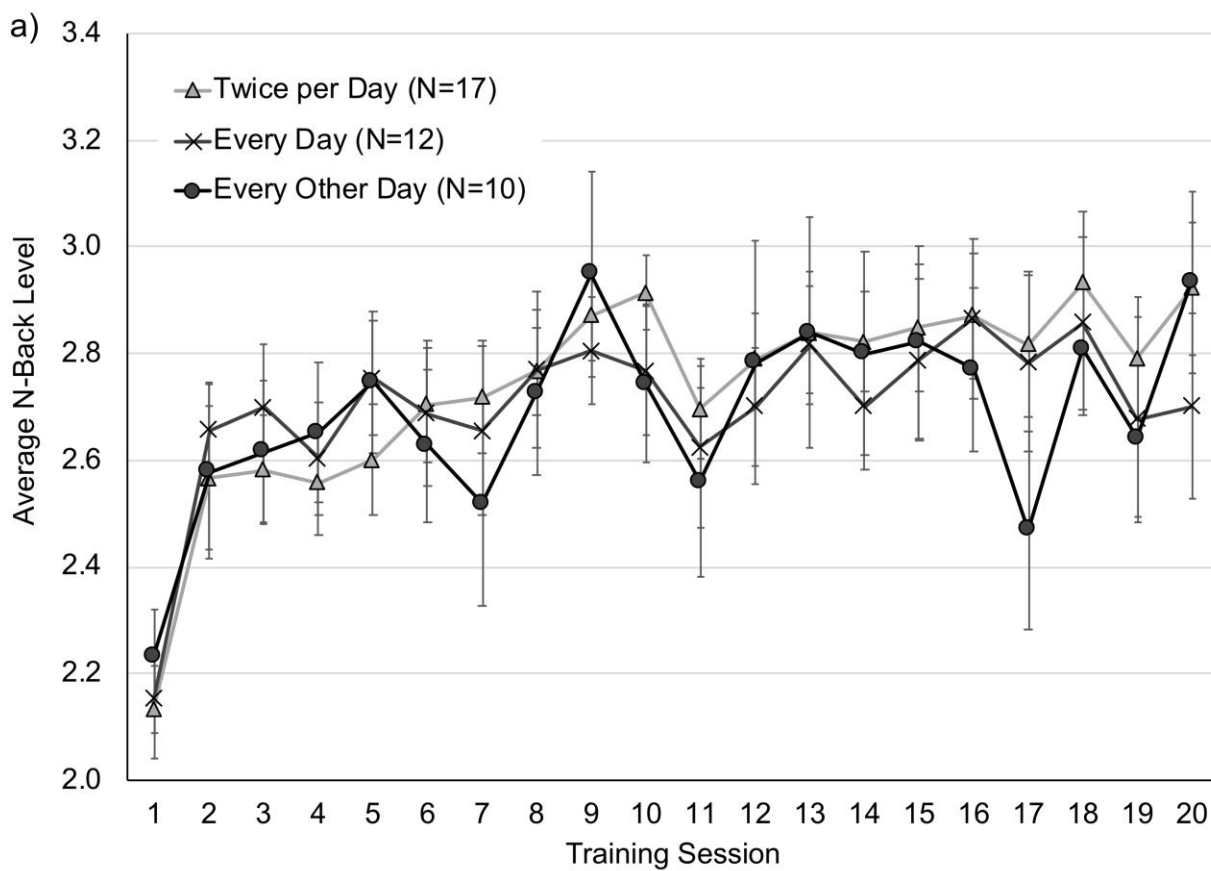
Figure 3a

Figure 3b

